

Neural Diet Tracker

A side project using Meta's TRIBE v2 to analyze your TikTok watch history. Build plan, scope, architecture, and milestones.

1. Project Summary

A local-first web app that ingests a user's TikTok data export, downloads the videos they watched, and runs each one through Meta FAIR's TRIBE v2 foundation model to predict the average human brain's cortical response. The output is a personal report — a “Spotify Wrapped for your attention” — showing predicted neural engagement curves, content-type clusters, and divergence between day-of-week or time-of-day viewing patterns.

The project demonstrates: real ML systems work (deploying a recent foundation model with GPU inference), end-to-end product engineering (data ingestion through visualization), and a sticky one-sentence pitch that survives an 8-second resume skim. Target audience for the deliverable is SWE internship recruiters at consumer-product companies (Meta, Snap, Roblox, Discord, Stripe).

2. Scope — What It Claims and What It Doesn't

The honest framing matters more than the code. Most projects in this category overclaim and get torn apart by anyone with a neuroscience background. The framing below is the one that survives scrutiny.

What the tool DOES measure

- bullet** Predicted BOLD fMRI response (cortical surface, 20,484 vertices) for each watched video, from the average of TRIBE v2's 720 training subjects.
- bullet** Predicted engagement in well-localized cortical regions: visual cortex, auditory cortex, language network, face-selective regions.
- bullet** Divergence and similarity between videos — clustering content by predicted neural fingerprint.
- bullet** Aggregate patterns over a user's watch history: time-of-day shifts, content-type breakdowns, session-level variation.

What it does NOT measure (do not claim these)

- bullet** Individual brain response — the model predicts an average, not the specific user's brain.
- bullet** Subcortical reward/addiction circuitry — nucleus accumbens, VTA, etc. are not in the output space.
- bullet** Cumulative effects, habituation, or neuroplasticity over time.
- bullet** Subjective states: aesthetic preference, confusion, cognitive overload, memory encoding, satisfaction.
- bullet** “How TikTok is rewiring your brain” or any rhetoric in that family.

Framing rule. Brand the tool as a “neural diet tracker” or “predicted neural engagement profiler.” Resist the temptation to call it an “addiction detector” or “cognitive overload analyzer” for marketing reasons. The boring framing is the one that holds up in technical interviews and on Twitter.

3. System Architecture

Local-first, single-user. No multi-tenant infrastructure, no auth, no cloud database. The user runs the app on their own machine (or a rented GPU) against their own TikTok export.

Component	Responsibility
Importer	Parses TikTok data export JSON; extracts watched-video URLs and metadata.
Downloader	Fetches videos via yt-dlp with rate limiting, retry, and dedup.
Preprocessor	Normalizes video to TRIBE's expected format (resolution, FPS, audio sample rate).
Inference service	FastAPI wrapper around TRIBE v2. Batch-processes videos on GPU; returns activation tensors.
Aggregator	Reduces raw activations to region-level metrics; computes summary statistics over time.
Storage	SQLite for metadata + on-disk Parquet for activation arrays.
Frontend	React/Next.js dashboard. Brain timeline visualization, cluster views, comparison mode.
Orchestrator	Job queue (Redis/RQ or just SQLite-based queue) coordinating download & rarr; preprocess & rarr; infer

4. Tech Stack

Layer	Choice	Why
ML model	TRIBE v2 (facebook/tribev2 on HF)	The core capability. CC BY-NC license — non-commercial only.
Inference runtime	PyTorch + CUDA	Standard. Requires GPU; Colab works, RunPod/Modal/Fly.io for hosted.
Backend	Python 3.11 + FastAPI	Clean async API, easy to spin up an inference service.
Video tooling	yt-dlp, ffmpeg	Robust downloading and preprocessing.
Queue	RQ (Redis) or SQLite-backed	Don't overbuild; single-user load.
Storage	SQLite + Parquet (pyarrow)	Metadata in SQL, activation tensors as Parquet on disk.
Frontend	Next.js + TypeScript + Tailwind	Modern, hireable stack.
Viz	D3.js + Three.js (brain mesh)	Brain heatmap is the hero visual; lean on Three.js for the 3D cortical surface.
Packaging	Docker compose for local; Modal for GPU	Local command local run; deployable GPU step is optional.

5. Build Phases

Five phases, designed so that each phase is independently demoable. If you run out of time at any phase boundary, you still have something shippable. Do not move on until the current phase is solid.

Phase 1 — Foundation (weekend 1)

Get TRIBE v2 running end-to-end on a single video.

- bullet Set up GPU environment (Colab Pro is the cheapest path; RunPod for ~\$0.40/hr).
- bullet Accept LLaMA 3.2 license on HuggingFace; download TRIBE v2 weights.
- bullet Run a single TikTok clip through the model. Verify output shape (20,484 vertices × T seconds).
- bullet Render a basic 3D brain heatmap in a Jupyter notebook using nilearn or pycortex. This is the proof-of-life.

Phase 2 — Data Pipeline (weekend 2)

Move from one video to a batch from a real TikTok export.

- bullet Parse the TikTok data export JSON (specifically the Activity / Video Browsing History).
- bullet Build a yt-dlp wrapper with retry, rate-limit handling, and dedup.
- bullet Add preprocessing: resize, re-encode, normalize audio. Cache aggressively — redownloading is slow.
- bullet Run inference in batches; persist results as Parquet keyed by video ID.
- bullet Wire SQLite schema: videos, watch_events, inference_runs.

Phase 3 — Aggregation & Insight Layer (weekend 3)

Turn raw activations into a small set of interpretable, defensible metrics.

- bullet Define region masks: visual cortex (V1–V4), auditory cortex, language network, FFA, VWFA.
- bullet For each video, compute time-series of mean activation per region.
- bullet Compute summary metrics: peak engagement, sustained engagement, region balance, novelty score.
- bullet Cluster videos by activation fingerprint (UMAP + HDBSCAN or simple k-means).
- bullet Aggregate across the user's history: time-of-day, day-of-week, weekly trend.

Phase 4 — Frontend (weekend 4–5)

The viz layer is the project. This is where most of the perceived quality lives.

- bullet Next.js app shell with three core views: Dashboard, Video Detail, Compare.
- bullet Dashboard: brain-timeline of the last 30 days, region breakdown, top clusters.
- bullet Video Detail: thumbnail, 3D brain heatmap (Three.js) playing in sync with the video.
- bullet Compare: side-by-side activation curves between any two videos or time periods.
- bullet Polish: dark mode default, typographic care, no Tailwind-template look.

Phase 5 — Polish & Deliverables (weekend 6)

Without this phase the project is invisible. Treat it as half the project.

- bullet 60-second demo video. No narration needed. Show: import → processing → the brain timeline. End frame is the comparison view.
- bullet README with: one-sentence pitch, GIF, install instructions, honest framing of what TRIBE measures.

bullet Blog post (Medium/Substack/personal site) articulating the design choices and the honest framing.

bullet Tag the repo, write a Twitter/Bluesky thread linking the demo, deploy a static landing page.

6. Engineering Challenges Worth Anticipating

bullet GPU memory. TRIBE v2 uses LLaMA 3.2-3B + V-JEPA2 + Wav2Vec-BERT. Expect ~16GB VRAM minimum. Batch carefully.

bullet t-dlp rate limits. TikTok aggressively rate-limits. Add jitter, rotate user agents, expect occasional failures — design for resumability.

bullet License compliance. TRIBE v2 is CC BY-NC. The project must stay non-commercial. State this clearly in the README and on the landing page.

bullet Out-of-distribution risk. TikTok content is closer to TRIBE's training data than most stimuli, but it's not identical. Be cautious with claims about absolute activation magnitudes; comparative claims are safer.

bullet Privacy. Watch history is sensitive. Local-first means the data never leaves the user's machine. Do not add an analytics SDK that exfiltrates anything.

bullet Reproducibility. Random seeds, model versions, preprocessing parameters should all be logged with each inference run.

7. Frontend Design Direction

The visual identity should be: scientific instrument meets consumer wrapped-style report. Think Linear's typography meeting NASA mission-control density meeting Spotify Wrapped's shareability. The 3D brain mesh is the hero element — everything else supports it.

bullet Dark mode default. Neural activation maps read better on dark backgrounds.

bullet Single accent color (warm, e.g. teal or amber) used sparingly for emphasis on activation peaks.

bullet Typography: a geometric sans (Inter, General Sans) for UI; a serif (Fraunces, Source Serif) for the narrative/"wrapped" section.

bullet Timeline scrubbing must feel responsive — this is the demo moment. Pre-render heatmap frames if needed.

bullet Avoid: rounded-2xl card grids, generic Tailwind components, anything that looks like a Vercel template.

8. Definition of Done

The project ships when all of the following are true:

bullet A friend with no setup help can clone the repo, follow the README, drop in their own TikTok export, and produce a report within an hour.

bullet The 3D brain visualization plays smoothly in sync with at least one demo video on a mid-range laptop.

bullet The demo video exists and is under 90 seconds.

bullet The blog post / writeup is published and the README links to it.

bullet The README contains explicit, prominent language about what TRIBE does and doesn't measure.

9. Optional Enhancement: Calibration Model

This addition shifts the project from “consumed an ML model” to “did ML on top of an ML model.” Highest-leverage single addition for portfolio strength. Adds roughly one weekend of work.

- bullet** For ~200 of your own watched videos, log behavioral signals: watch duration, rewatched yes/no, liked, saved, scrolled-past-quickly.
- bullet** Extract TRIBE-derived features per video: region-level summary statistics from the inference output.
- bullet** Train a small model (gradient boosting; XGBoost or LightGBM) that predicts each behavioral signal from the TRIBE features.
- bullet** Report R^2 / AUC on held-out videos. Be honest about what generalizes and what doesn't.
- bullet** Add a “Predicted personal engagement” score to the dashboard, based on the calibrated model.

10. Realistic Timeline

Assuming ~10–15 focused hours per week on top of coursework. Compress if winter or summer break is available.

Week	Milestone	Risk
1	Phase 1: single-video TRIBE inference works locally.	GPU setup friction.
2	Phase 2: pipeline processes a full TikTok export.	yt-dlp / rate limits.
3	Phase 3: aggregation + clustering layer produces clean metrics.	Defining region masks correctly.
4–5	Phase 4: full frontend with brain visualization.	Three.js learning curve; perf tuning.
6	Phase 5: demo video, README, blog post, launch.	Skipping this phase is the #1 failure mode.
7 (optional)	Calibration model on personal data.	Sample size.

If you remember one thing. The single highest-leverage decision in this whole plan is whether you actually do Phase 5. A polished demo video and a clear writeup are worth more for internship recruiting than any additional technical sophistication you could add to the code. Phase 5 is not optional.